

RECURSION

“YOUR **FUTURE** IS CREATED BY WHAT
YOU DO **{ TODAY }**
NOT ~~TOMORROW~~”

PHASE I

- ✓ *Introduction to C++*
- ✓ *Loops and Pattern Printing*
- ✓ *Arrays / 2D Arrays*
- ✓ *Strings*
- ✓ *Sorting and Searching*

PHASE II

- ✓ *Time and Space Complexity*
- ✓ *Sets and Maps*
- ✓ *Prefix Sums & Contribution*
- ✓ *Sliding Window & Two Pointers*
- ✓ *Number Theory Basics*
- ✓ *Bit Manipulation*

PHASE III

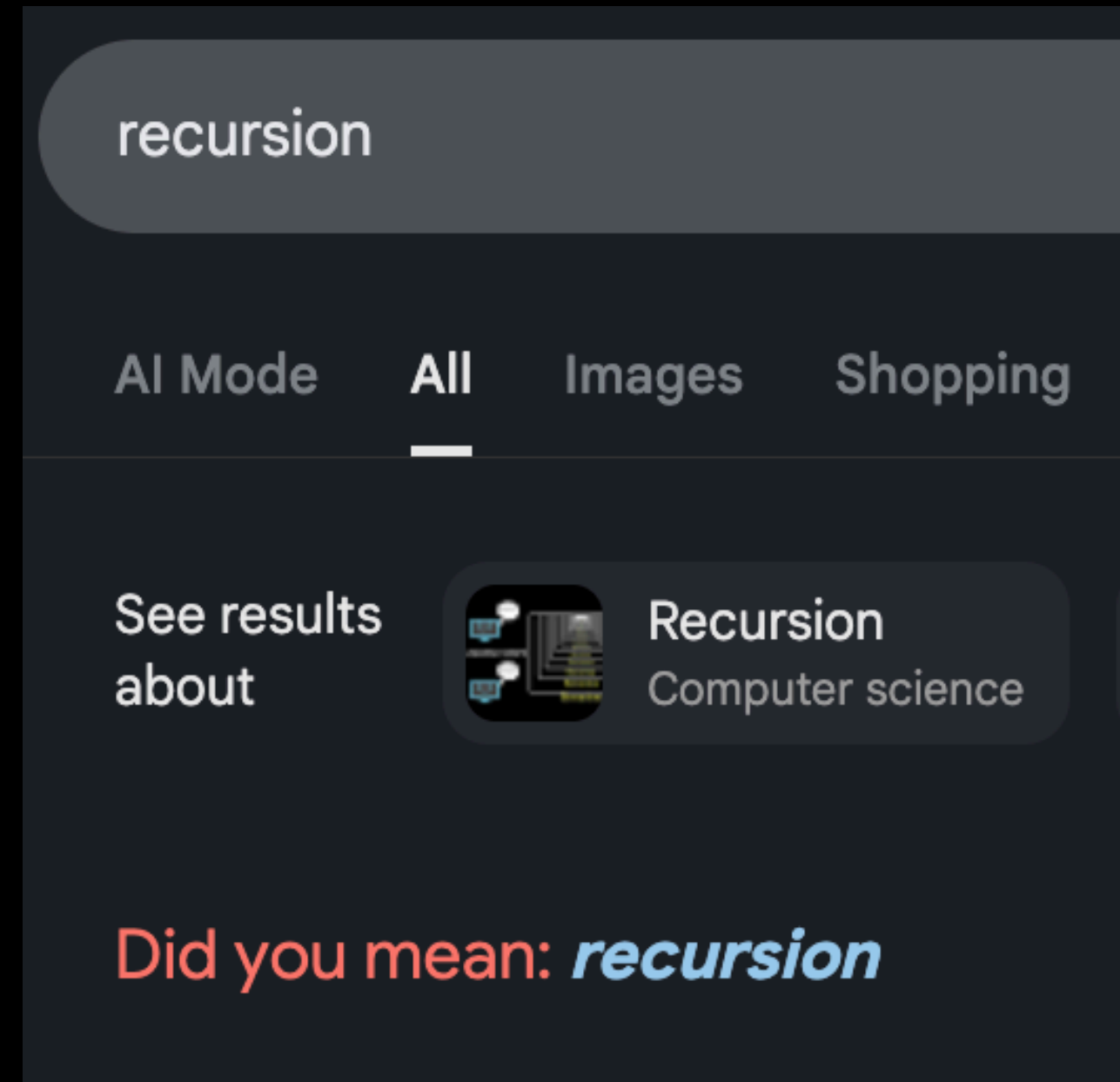
Recursion
Backtracking
Linked List
Stack / Queue / Deque
Binary Tree / BST
Priority Queue / Heap

PHASE IV

Greedy
Graphs
Dynamic Programming
Trie
Segment Tree / Ordered Set

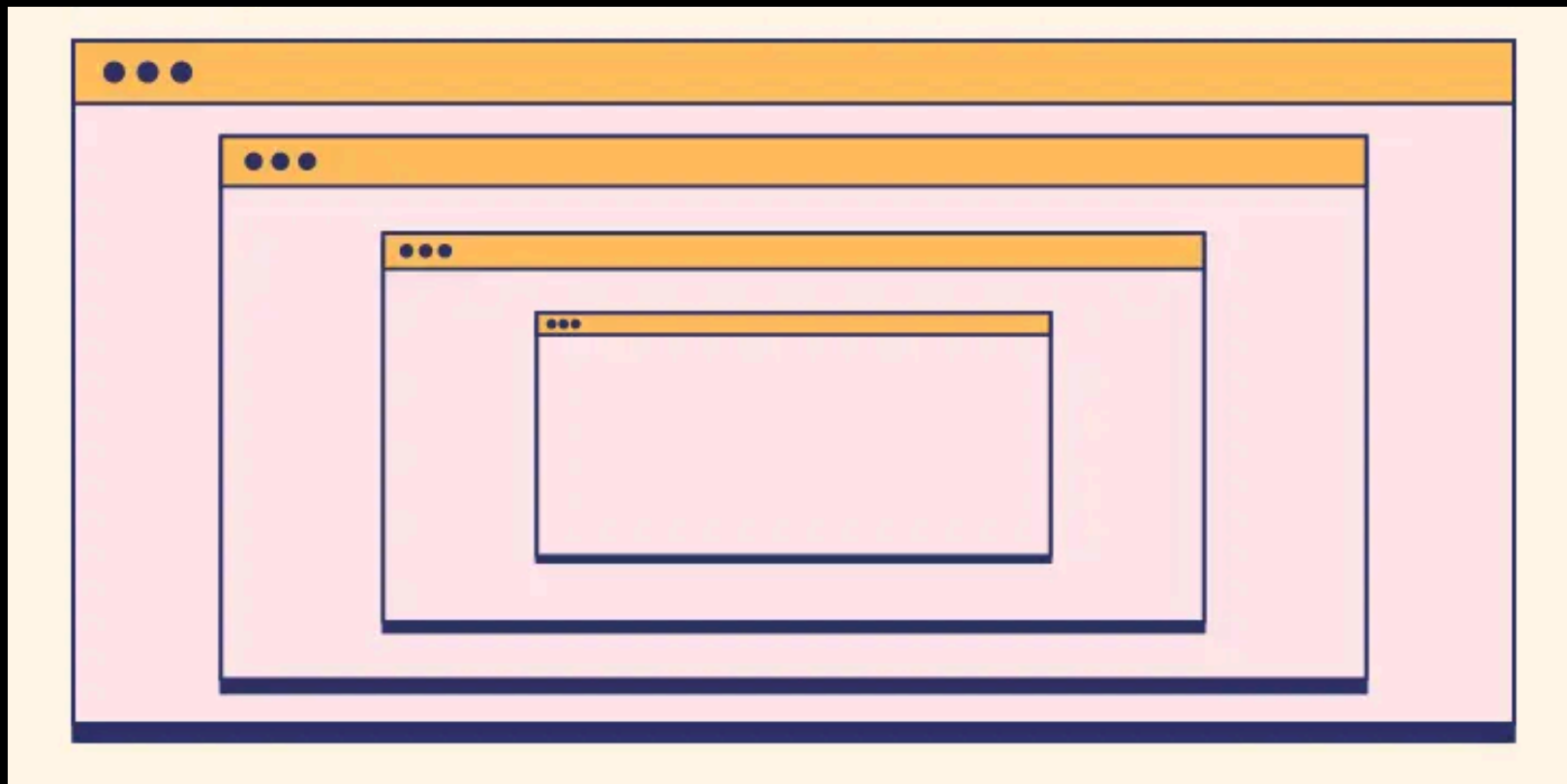
WHAT IS RECURSION?

Function calling itself



WHAT IS RECURSION?

Function calling itself with a **smaller input** till a stopping point (**base case**).



PRINT DECREASING

Given an integer n , print n to 1 without using any loop.

$n = 5$

5

4

3

2

1

$n = 6$

6

5

4

3

2

1

$n = 7$

7

6

5

4

3

2

1

THREE **GOLDEN** RULES TO SOLVE ANY RECURSION PROBLEM

- 1) Define a **clear cut responsibility** for the recursive function.
- 2) Break it down recursively and **Believe / Trust / Have a leap of faith**, that this function will work for smaller inputs automatically.
- 3) Figure out the **base case**

PRINT INCREASING

Given an integer n , print 1 to n without using any loop.

$n = 5$

1

2

3

4

5

$n = 6$

1

2

3

4

5

6

$n = 7$

1

2

3

4

5

6

7

PRINT ZIG ZAG

Given an integer n , print zig zag sequence. (n to 1 and then back to n)

$n = 5$

5
4
3
2
1
2
3
4
5

$n = 4$

4
3
2
1
2
3
4

$n = 1$

1

PRINT NUMBER IN REVERSE

Given an integer **n**, print the digits of n in reverse order.

n = 12349

94321

n = 1950

0591

RETURNING ANSWER PROBLEMS

FACTORIAL

Given an integer N , compute $N!$ using recursion.

Note: You must solve it using **recursion**.

The factorial of a non-negative integer N is defined as $N! = \prod_{i=1}^N i$, with $0! = 1$.

Input

A single integer N ($0 \leq N \leq 20$).

Output

Print $N!$.

Example

input
5
output
120

SUM FROM 1 TO N

Given an integer N , compute the sum $1 + 2 + 3 + \dots + N$ using recursion.

Let $S(N) = 1 + 2 + 3 + \dots + N$.

Input

A single integer N ($1 \leq N \leq 10^5$).

Output

Print $S(N)$.

Example

input
5
output
15

POWER

Given two integers X and N , compute X^N using recursion.

Input

Two integers X and N ($1 \leq X \leq 10$, $1 \leq N \leq 18$).

Output

Print the value of X^N .

Example

input
2 3
output
8

SUM OF DIGITS

Given an integer N , find the sum of its digits.

Input

A single integer N ($0 \leq N \leq 10^{18}$).

Output

Print the sum of digits of N .

Example

input
1234
output
10

FIBONACCI

Given an integer N , print the N -th Fibonacci number.

The Fibonacci sequence is defined as:

$F_1 = 0$, $F_2 = 1$, and for $N \geq 3$,

$$F_N = F_{N-1} + F_{N-2}$$

Input

A single integer N ($1 \leq N \leq 20$).

Output

Print F_N .

Example

input
2
output
1

RECURSION

- *What is Recursion?*
- *Printing Problems: Print Decreasing, Print Increasing, Print Zig Zag, Print Number in Reverse*
- *Returning Answer: Factorial, Sum of First N Natural Numbers, Power, Sum of Digits, Nth Fibonacci*
- *Miscellaneous: Tower of Hanoi*
- *Arrays / Strings + Recursion*